Analysis report for sample: 457955316cee75bbda8e0bee0925598d

This report has been generated by a set of tools for semi-supervised behavioral analysis of executables.

More information about these reports at **http://www.vallejo.cc**.

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Report classification: 3060 (indicates the level of interest of the report, based on the amount of information collected about the sample).

Index

- Sample
- AV detections
- Original source url
- Virustotal url
- Yara matches
- Snapshot
- Threads tree
- Running processes
- Most interesting actions
- Most interesting strings
- Involved hosts
- Dns queries
- Connections and readable network traffic
- Threads behavioural information
- Malware processes' network events
- Sample's unpacked / injected modules

Sample

- md5: 457955316cee75bbda8e0bee0925598d

AV detections

- Microsoft:
- Kaspersky: Trojan-Ransom.Win32.Zerber.xh
- Symantec:
- Malwarebytes: Ransom.Cerber

Original source url

-

Virustotal url

- https://virustotal.com/es/file/ebb16fa53de04930dcbbdfa80141322071248f4134eeb2b923c85d9716907263/analysis

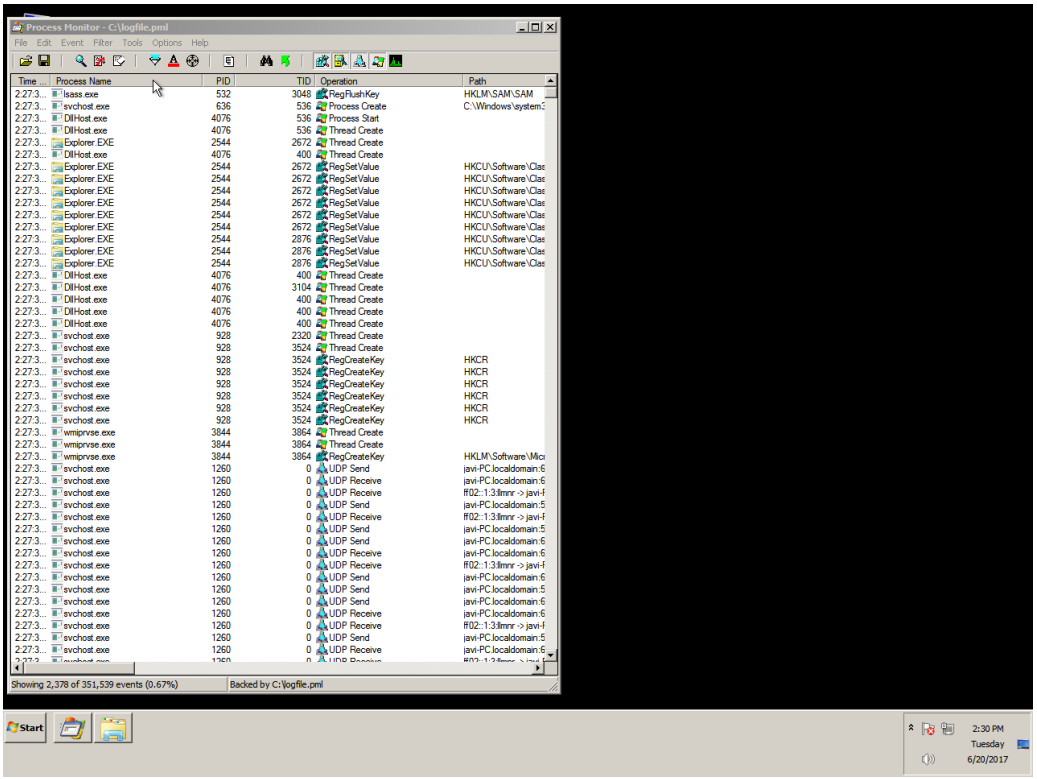
Yara matches

The following yara rules have matched injected or unpacked modules's code or data areas.

- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- HasRichSignature (Rich Signature Check)
- DebuggerHiding__Active ()
- anti_dbg (Checks if being debugged)
- screenshot (Take screenshot)
- keylogger (Run a keylogger)
- win_mutex (Create or check mutex)
- win_registry (Affect system registries)
- win_files_operation (Affect private profile)
- win_hook (Affect hook table)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- maldoc_getEIP_method_1 ()
- contentis_base64 (This rule finds for base64 strings)
- GenerateTLSClientHelloPacket_Test ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- Str_Win32_Http_API (Match Windows Http API call)
- ASProtect_v132 ()
- HasOverlay (Overlay Check)
- HasDebugData (DebugData Check)
- DebuggerCheck__QueryInfo ()
- Check_Dlls ()
- Check_Qemu_Description ()
- Check_Qemu_DeviceMap ()
- Check_VBox_Description ()
- Check_VBox_DeviceMap ()
- Check_VBox_Guest_Additions ()

- Check_VBox_VideoDrivers ()
- Check_VMWare_DeviceMap ()
- Check_VmTools ()
- Check_Wine ()
- vmdetect (Possibly employs anti-virtualization techniques)
- Check_OutputDebugStringA_iat ()
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_http (Communications over HTTP)
- escalate_priv (Escalade priviledges)
- win_token (Affect system token)
- vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)
- Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
- MD5_Constants (Look for MD5 constants)
- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

Snapshot


Process Monitor - C:\logfile.pml
File Edit Event Filter Tools Options Help
Time ... Process Name PID TID Operation Path
2:27:3... lsass.exe 532 3048 RegFlushKey HKLM\SAM\SAM
2:27:3... svchost.exe 636 536 Process Create C:\Windows\system3
2:27:3... DllHost.exe 4076 536 Process Start
2:27:3... DllHost.exe 4076 536 Thread Create
2:27:3... Explorer.EXE 2544 2672 Thread Create
2:27:3... DllHost.exe 4076 400 Thread Create
2:27:3... Explorer.EXE 2544 2672 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2672 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2672 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2672 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2672 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2672 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2876 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2876 RegSetValue HKCU\Software\Clas
2:27:3... Explorer.EXE 2544 2876 RegSetValue HKCU\Software\Clas
2:27:3... DllHost.exe 4076 400 Thread Create
2:27:3... DllHost.exe 4076 3104 Thread Create
2:27:3... DllHost.exe 4076 400 Thread Create
2:27:3... DllHost.exe 4076 400 Thread Create
2:27:3... DllHost.exe 4076 400 Thread Create
2:27:3... svchost.exe 928 2320 Thread Create
2:27:3... svchost.exe 928 3524 Thread Create
2:27:3... svchost.exe 928 3524 RegCreateKey HKCR
2:27:3... svchost.exe 928 3524 RegCreateKey HKCR
2:27:3... svchost.exe 928 3524 RegCreateKey HKCR
2:27:3... svchost.exe 928 3524 RegCreateKey HKCR
2:27:3... svchost.exe 928 3524 RegCreateKey HKCR
2:27:3... svchost.exe 928 3524 RegCreateKey HKCR
2:27:3... wmiprvse.exe 3844 3864 Thread Create
2:27:3... wmiprvse.exe 3844 3864 Thread Create
2:27:3... wmiprvse.exe 3844 3864 RegCreateKey HKLM\Software\Micr
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Receive javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Receive ff02::1:3:llmnr -> javi-f
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:5
2:27:3... svchost.exe 1260 0 UDP Receive ff02::1:3:llmnr -> javi-f
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:5
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Receive javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Receive ff02::1:3:llmnr -> javi-f
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:5
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Receive javi-PC.localdomain:6
2:27:3... svchost.exe 1260 0 UDP Receive ff02::1:3:llmnr -> javi-f
2:27:3... svchost.exe 1260 0 UDP Send javi-PC.localdomain:5
2:27:3... svchost.exe 1260 0 UDP Receive javi-PC.localdomain:6
Showing 2,378 of 351,539 events (0.67%)
Backed by C:\logfile.pml
Start
2:30 PM
Tuesday
6/20/2017

Threads tree

The following tree represents sample's threads. T<index> is an alias for sample's threads (numeration is done in the order of threads creation). P<index> is an alias for processes owning sample's threads.

Running processes

Here it is the list of running processes after sample execution.

Processes with sample' threads

Other processes

- Name System
- Name smss.exe
- Name csrss.exe
- Name wininit.exe
- Name csrss.exe
- Name winlogon.exe
- Name services.exe
- Name lsass.exe
- Name lsm.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name svchost.exe
- Name audiodg.exe
- Name TrustedInstaller.exe
- Name svchost.exe
- Name svchost.exe
- Name spoolsv.exe
- Name svchost.exe
- Name svchost.exe
- Name vmtoolse.exe
- Name svchost.exe
- Name dllhost.exe
- Name msdtc.exe
- Name sppsvc.exe
- Name taskhost.exe
- Name dwm.exe
- Name explorer.exe
- Name vmtoolse.exe
- Name SearchIndexer.exe
- Name svchost.exe
- Name wmpnetwk.exe
- Name WmiPrvSE.exe
- Name chrome.exe
- Name SearchFilterHost.exe
- Name taskhost.exe
- Name SearchProtocolHost.exe
- Name pxxx.exe
- Name explorer.exe
- Name explorer.exe
- Name cebo.exe
- Name icardagt.exe
- Name cmd.exe
- Name conhost.exe
- Name taskkill.exe
- Name vssadmin.exe

- Name conhost.exe
- Name WmiPrvSE.exe
- Name VSSVC.exe
- Name svchost.exe
- Name cmd.exe
- Name conhost.exe
- Name icardagt.exe
- Name vssadmin.exe
- Name conhost.exe
- Name cmd.exe
- Name conhost.exe
- Name icardagt.exe
- Name cmd.exe
- Name conhost.exe
- Name icardagt.exe
- Name explorer.exe
- Name explorer.exe

Most interesting actions

The following list it's a collection of the most interesting events captured. This list is ordered by the score assigned to the event. In the section "Threads behavioural information" it's possible to find all the actions performed by each sample's thread ordered chronologically.

Most interesting strings

The following list it's a collection of the most interesting strings found in the sample's modules code or data. A complete list of strings for each sample's module can be found in the section "Sample's unpacked / injected modules."

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe
- path | path unit | wellknown extension: C:\\Users\\javi\\Downloads\\cebo.exe
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer
- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI
- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- path | path unit: C:\\Users\\javi\\Downloads\\
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA
- api | wininet: HttpSendRequestA

- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem
- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey
- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW
- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect
- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW
- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW
- api | advapi32: SetKernelObjectSecurity

- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree
- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId
- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA
- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW
- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW
- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW
- api | kernel32: CreateFileMappingW

- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary
- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW
- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW
- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW
- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode
- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

Involved hosts

- 192.168.133.1:5355
- 192.168.133.232:50102
- 192.168.133.232:62075

Dns queries

Connections and readable network traffic

The analysis environment captures network traffic happened while sample execution. It classifies the traffic and tries to extract the most significant parts from the communications data.

In this section it's possible to find parts of the communication data that are easier to be read.

- **udp 192.168.133.1:5355 ---> 192.168.133.232:62075**

  in-addr[...]arpa[...]in-addr

- **udp 192.168.133.1:5355 ---> 192.168.133.232:50102**

  arpa

Threads behavioural information

In this section it's possible to find information about sample's threads, such as the actions performed by each sample's thread ordered chronologically.

Malware processes' network events

The analysis environment tries to capture and collect network actions performed by sample's threads.

Sample's unpacked / injected modules

In this section it's possible to find information about sample's modules, such as the rich signatures, strings, yara rules matching the module,...

The analysis environment performs an heuristic search trying to find sample's injected or unpacked modules in memory, dumping and analyzing them.

• Module 1 (probably unpacked / injected by the sample)

• Module 1 rich signatures

• 44616e530000000000000000000000000c30f0f0002000000c30f5f0001000000c30f5d00020000001b9dab000100000009788300030000000978930021000000000001007c0100001b9daf00510000001b9d9d00

• Module 1 yara matches

• GenerateTLSClientHelloPacket_Test ()
• Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
• Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
• Str_Win32_Internet_API (Match Windows Inet API call)
• Str_Win32_Http_API (Match Windows Http API call)
• ASProtect_v132 ()
• Microsoft_Visual_Cpp_v50v60_MFC ()
• IsPE32 ()
• IsWindowsGUI ()
• HasOverlay (Overlay Check)
• HasDebugData (DebugData Check)
• HasRichSignature (Rich Signature Check)
• DebuggerCheck__QueryInfo ()
• Check_Dlls ()
• Check_Qemu_Description ()
• Check_Qemu_DeviceMap ()
• Check_VBox_Description ()
• Check_VBox_DeviceMap ()
• Check_VBox_Guest_Additions ()
• Check_VBox_VideoDrivers ()
• Check_VMWare_DeviceMap ()
• Check_VmTools ()
• Check_Wine ()
• vmdetect (Possibly employs anti-virtualization techniques)
• Check_OutputDebugStringA_iat ()
• anti_dbg (Checks if being debugged)
• inject_thread (Code injection with CreateRemoteThread in a remote process)
• network_http (Communications over HTTP)
• escalate_priv (Escalade priviledges)
• win_mutex (Create or check mutex)
• win_registry (Affect system registries)
• win_token (Affect system token)
• win_files_operation (Affect private profile)
• vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)
• without_attachments (Rule to detect the no presence of any attachment)
• without_images (Rule to detect the no presence of any image)
• without_urls (Rule to detect the no presence of any url)
• contentis_base64 (This rule finds for base64 strings)
• Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
• MD5_Constants (Look for MD5 constants)

- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

• Module 1 strings

• Module 1 most interesting strings

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer
- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI
- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA

- api | wininet: HttpSendRequestA
- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem
- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey
- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW
- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect
- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW
- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW

- api | advapi32: SetKernelObjectSecurity
- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree
- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId
- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA
- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW
- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW
- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW

- api | kernel32: CreateFileMappingW
- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary
- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW
- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW
- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW
- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode
- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

• Module 1 other strings

• Printers\\Defaults\\%s
• CERBER_KEY_PLACE
• NoDrives
• open
• EnableLUA
• /set {default} bootstatuspolicy ignoreallfailures
• /set {default} recoveryenabled no
• shadowcopy delete
• %s\\wbem\\wmic.exe
• delete shadows /all /quiet
• COMSPEC
• Installed
• PT0S
• PT1M
• 2000-01-01T00:00:00
• Component_02
• SeDebugPrivilege
• CERBER_BODY_PLACE
• CERBER_EVALUATED_CORE_PROTECTION_EVENT
• CERBER_CORE_PROTECTION_MUTEX
• \\x*x.exe
• update
• setup
• install
• sol.exe
• spider
• calc
• cmd.exe
• freecell
• mshearts
• ntbackup
• route
• task
• telnet
• winmine
• ping
• netsh
• notepad
• logoff
• reset
• msiexec
• winlogon
• disk
• lookup
• attrib
• find
• ftp.exe
• netstat
• write
• copy
• *.exe
• eshell32.dll
• "%s\\explorer.exe"

- %COMSPEC%
- runas
- Component_01
- Component_00
- files.txt
- Modified at %s %s
- ekernel32.dll
- vmmouse.sys
- SOFTWARE\\VMware, Inc.\\VMware Tools
- WMWARE
- VMWARE
- VBoxMouse.sys
- \\drivers\\
- SOFTWARE\\Oracle\\VirtualBox Guest Additions
- VIRTUALBOX
- VBOX
- QEMU
- HARDWARE\\DEVICEMAP\\Scsi\\Scsi Port 0\\Scsi Bus 0\\Target Id 0\\Logical Unit Id 0
- Identifier
- VideoBiosVersion
- PARALLELS
- HARDWARE\\Description\\System
- SystemBiosVersion
- multi_pot.exe
- hookexplorer.exe
- proc_analyzer.exe
- scktool.exe
- sniff_hit.exe
- sysanalyzer.exe
- idag.exe
- ollydbg.exe
- dumpcap.exe
- wireshark.exe
- \\\\.\\cv2k1
- \\sandbox\\
- \\sand-box\\
- Frz_State
- test_item.exe
- Ajjj
- Ajjj
- Ajjj
- U9f:f?6
- xo*2
- _6}\\
- 6lpFn
- zj-o
- }!)E
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- $@1'!
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- l9}¡^
- .n=4H6
- oqJ=
- U*+3
- !O=B-om
- r|5U

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- Ysxa
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- #+3`
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- LHYId
- 09sr
- IM@;
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- ~1dqf
- *2,+
- lYuh
- uQkY
- -&PA;
- ^PAI
-]LHo
- iPGM
- fAjN#
- ed6T?H`:
- \\3AA3$
- H7OXJgM0
- Y9qaA
- z3yV
- }aRY
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- #TeoD
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- D$ a
- D$ aX
- VerQueryValueW
- GetFileVersionInfoW
- StrToInt64ExA
- PathSkipRootW
- PathUnquoteSpacesW
- PathCombineW
- CoInitialize
- NETAPI32.dll
- OLEAUT32.dll
- SHELL32.dll
- ole32.dll
- USER32.dll
- ADVAPI32.dll
- KERNEL32.dll
- ShellExecuteW
- SHChangeNotify
- CoInitializeSecurity
- SHGetFolderPathW
- ShellExecuteA
- CoInitializeEx
- ShellExecuteExW
- CoUninitialize
- WS2_32.dll
- imagehlp.dll
- CheckSumMappedFile
- WNetEnumResourceW
- WNetOpenEnumW
- WNetCloseEnum
- VERSION.dll
- GetFileVersionInfoA
- GetFileVersionInfoSizeA
- VerQueryValueA
- GetFileVersionInfoSizeW
- PathMatchSpecW
- PathRemoveExtensionW

- StrCmpNIA
- PathFindFileNameW
- StrStrIA
- StrChrIA
- WININET.dll
- lA-*
- c"5o
- T1`m
- 4d(2d
- lDx3
- ?U Up
- Ydm3
- o38Fg$
- h6J-
- 4w%sd
- WV_i
- whitelist
- folders
- new_extension
- file_body
- file_extension
- files
- files_name
- help_files
- network
- debug
- activity
- language
- country
- vmware
- check
- countries
- property_name
- ip_geo
- languages
- blacklist
- kernel32.dll
- {COUNT_FILES}
- %02X%02X%02X%02X%02X%02X
- {MD5_KEY}
- {STOP_REASON}
- {OS}
- %05x
- {PARTNER_ID}
- {COUNTRY}
- {IS_X64}
- {IS_ADMIN}
- data_finish
- data_start
- knock
- port
- %02x
- timeout
- statistics
- servers
- %02X%02X%02X%02X%02X%02X%05X%03X
- %02x%02x%02x%02x%02x%02x%05xcerber

- open
- Stop reason: %s
- SHCreateItemFromParsingName
- requireAdministrator
- requestedExecutionLevel
- true
- autoElevate
- global_public_key
- autostart
- rsa_key_size
- min_file_size
- max_blocks
- max_block_size
- multithread
- close_process
- wine_get_unix_file_name
- Fcerber
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- j X+E
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- j1h@
- Pha-A
- -uYh
- jxX+E
- j8X+E
- j@X+E
- E8@P
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- E(+E
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- hDo)
- h`5y
- u7j
- uYjY
- P8hT
- t6hH
- h<I@
- hjA@
- t5h`
- t0hT
- u:h|
- SVW2
- @.data
- `.rdata
- .text

- Rich
- !This program cannot be run in DOS mode.
- \\??\\
- *%S*
- %S%S
- /d /c taskkill /t /f /im "%s" > NUL & ping -n 1 127.0.0.1 > NUL & del "%s" > NUL
- "%s\\%s"
- shell.%s
- %s\\%S
- %s\\%s
- /d /c start "" "%s"
- Encrypting file %s...
- %s\\%s%S
- vmhgfs.sys
- VEN_%x
- \\\\.\\NPF_NdisWanIp
- \\cwsandbox\\
- dbghelp.dll
- dir_watch.dll
- sbiedll.dll
- jjjj
- jjjjjj
- jjjjjj
- Bjjjj
- jjjj
- jjjjj
- Ajjjjj
- jjjj
- jjjj
- jjjj
- PADPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDIN GXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDIN GPADDINGXXPADDINGPADD
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-
- PPPP
- ``YYYYYYYYYYY
- }YPPPPYYYYa
- YYYYYYYYYYYY
- YY`}
- YYYYY
- StrChrA
- StrPBrkA
- ntdll.dll
- MPR.dll
- SHLWAPI.dll
- StrCmpNW
- StrChrW
- StrCmpIW
- StrSpnA
- StrCpyNW
- StrCmpNIW
- StrStrIW
- CRYPT32.dll
- $@RSDS
- gz|z
- K_VI
- Hx]k
- syFbg
- .vbs
- {PC_ID}
- Sending stat %s, %s
- %[^/]%[/]%d
- %c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c
- "%s"
- encrypt
- /-Rb
- Mb=Lk
- NKagj(h
- !Q\t@
- t+@;
- UQPXY]Y[
- ;t$(v(
-]_^[

-]_^[
- t?xH
- QQQSVW
- QQVW
- YYPj@j
- QQVh
- hSv3_j
- hSv3_j
- hEX&vj;ht
- nj-h
- hSv3_j
- Vj\th
- hSv3_j
- QQQSVW
- QQQSVW
- QQQSVW
- QQQSVW
- QQQQSVW
- j Y+
- PhSv3_j
- QQQQSVW
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- QZ^&
- Ph'c
- PhSv3_j
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- QQQQSVW
- YYh(
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- YYPh
- YYPj
- SVW
- ;HTw|j
- t?hT
- Phb$@
- crbr
- @tSj
- .rsrc

• Module 2 (probably unpacked / injected by the sample)

• Module 2 rich signatures

- 44616e530000000000000000000000000c30f0f0002000000c30f5f0001000000c30f5d000200000001b9dab000100000009788300030000000978930021000000000001007c0100001b9daf00510000001b9d9d00

• Module 2 yara matches

- GenerateTLSClientHelloPacket_Test ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- Str_Win32_Http_API (Match Windows Http API call)
- ASProtect_v132 ()
- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasDebugData (DebugData Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- Check_Dlls ()
- Check_Qemu_Description ()
- Check_Qemu_DeviceMap ()
- Check_VBox_Description ()
- Check_VBox_DeviceMap ()
- Check_VBox_Guest_Additions ()
- Check_VBox_VideoDrivers ()
- Check_VMWare_DeviceMap ()
- Check_VmTools ()
- Check_Wine ()
- vmdetect (Possibly employs anti-virtualization techniques)
- Check_OutputDebugStringA_iat ()
- anti_dbg (Checks if being debugged)
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_http (Communications over HTTP)
- escalate_priv (Escalade priviledges)
- win_mutex (Create or check mutex)
- win_registry (Affect system registries)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
- MD5_Constants (Look for MD5 constants)
- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

• Module 2 strings

• Module 2 most interesting strings

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe

- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer
- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI
- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA
- api | wininet: HttpSendRequestA
- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem
- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey

- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW
- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect
- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW
- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW
- api | advapi32: SetKernelObjectSecurity
- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree
- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId

- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA
- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW
- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW
- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW
- api | kernel32: CreateFileMappingW
- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary
- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW

- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW
- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW
- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode
- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

• Module 2 other strings

- Printers\\Defaults\\%s
- CERBER_KEY_PLACE
- NoDrives
- open
- EnableLUA
- /set {default} bootstatuspolicy ignoreallfailures
- /set {default} recoveryenabled no
- shadowcopy delete
- %s\\wbem\\wmic.exe
- delete shadows /all /quiet
- COMSPEC
- Installed
- PT0S

- PT1M
- 2000-01-01T00:00:00
- Component_02
- SeDebugPrivilege
- CERBER_BODY_PLACE
- CERBER_EVALUATED_CORE_PROTECTION_EVENT
- CERBER_CORE_PROTECTION_MUTEX
- \\x*x.exe
- update
- setup
- install
- sol.exe
- spider
- calc
- cmd.exe
- freecell
- mshearts
- ntbackup
- route
- task
- telnet
- winmine
- ping
- netsh
- notepad
- logoff
- reset
- msiexec
- winlogon
- disk
- lookup
- attrib
- find
- ftp.exe
- netstat
- write
- copy
- *.exe
- eshell32.dll
- "%s\\explorer.exe"
- %COMSPEC%
- runas
- Component_01
- Component_00
- files.txt
- Modified at %s %s
- ekernel32.dll
- vmmouse.sys
- SOFTWARE\\VMware, Inc.\\VMware Tools
- WMWARE
- VMWARE
- VBoxMouse.sys
- \\drivers\\
- SOFTWARE\\Oracle\\VirtualBox Guest Additions
- VIRTUALBOX
- VBOX
- QEMU

- HARDWARE\\DEVICEMAP\\Scsi\\Scsi Port 0\\Scsi Bus 0\\Target Id 0\\Logical Unit Id 0
- Identifier
- VideoBiosVersion
- PARALLELS
- HARDWARE\\Description\\System
- SystemBiosVersion
- multi_pot.exe
- hookexplorer.exe
- proc_analyzer.exe
- scktool.exe
- sniff_hit.exe
- sysanalyzer.exe
- idag.exe
- ollydbg.exe
- dumpcap.exe
- wireshark.exe
- \\\\.\\cv2k1
- \\sandbox\\
- \\sand-box\\
- Frz_State
- test_item.exe
- Ajjj
- Ajjj
- Ajjj
- U9f:f?6
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- 6lpFn
- zj-o
- }!)E
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- .n=4H6
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- #b$52S
- #+3`
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- *2,+
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- uQkY
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- H7OXJgM0
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- D$ a
- D$ aX
- VerQueryValueW
- GetFileVersionInfoW
- StrToInt64ExA
- PathSkipRootW
- PathUnquoteSpacesW
- PathCombineW
- CoInitialize
- NETAPI32.dll
- OLEAUT32.dll
- SHELL32.dll
- ole32.dll
- USER32.dll
- ADVAPI32.dll
- KERNEL32.dll
- ShellExecuteW
- SHChangeNotify
- CoInitializeSecurity
- SHGetFolderPathW
- ShellExecuteA
- CoInitializeEx
- ShellExecuteExW
- CoUninitialize
- WS2_32.dll
- imagehlp.dll
- CheckSumMappedFile
- WNetEnumResourceW
- WNetOpenEnumW
- WNetCloseEnum
- VERSION.dll
- GetFileVersionInfoA
- GetFileVersionInfoSizeA
- VerQueryValueA
- GetFileVersionInfoSizeW
- PathMatchSpecW
- PathRemoveExtensionW
- StrCmpNIA
- PathFindFileNameW
- StrStrIA
- StrChrIA
- WININET.dll
- lA-*
- c"5o
- T1`m
- 4d(2d
- IDx3
- ?U Up
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- o38Fg$
- h6J-
- 4w%sd
- WV_i
- whitelist

- folders
- new_extension
- file_body
- file_extension
- files
- files_name
- help_files
- network
- debug
- activity
- language
- country
- vmware
- check
- countries
- property_name
- ip_geo
- languages
- blacklist
- kernel32.dll
- {COUNT_FILES}
- %02X%02X%02X%02X%02X%02X
- {MD5_KEY}
- {STOP_REASON}
- {OS}
- %05x
- {PARTNER_ID}
- {COUNTRY}
- {IS_X64}
- {IS_ADMIN}
- data_finish
- data_start
- knock
- port
- %02x
- timeout
- statistics
- servers
- %02X%02X%02X%02X%02X%02X%05X%03X
- %02x%02x%02x%02x%02x%02x%05xcerber
- open
- Stop reason: %s
- SHCreateItemFromParsingName
- requireAdministrator
- requestedExecutionLevel
- true
- autoElevate
- global_public_key
- autostart
- rsa_key_size
- min_file_size
- max_blocks
- max_block_size
- multithread
- close_process
- wine_get_unix_file_name
- Fcerber

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- j X+E
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- Pha-A
- -uYh
- jxX+E
- j8X+E
- j@X+E
- E8@P
- E8@P
- E(+E
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- h`5y
- u7j
- uYjY
- P8hT
- t6hH
- h<I@
- hjA@
- t5h`
- t0hT
- u:h|
- SVW2
- @.data
- `.rdata
- .text
- Rich
- !This program cannot be run in DOS mode.
- \\??\\
- *%S*
- %S%S
- /d /c taskkill /t /f /im "%s" > NUL & ping -n 1 127.0.0.1 > NUL & del "%s" > NUL
- "%s\\%s"
- shell.%s
- %s\\%S
- %s\\%s
- /d /c start "" "%s"
- Encrypting file %s...
- %s\\%s%S
- vmhgfs.sys
- VEN_%x
- \\\\.\\NPF_NdisWanIp
- \\cwsandbox\\

• dbghelp.dll
• dir_watch.dll
• sbiedll.dll
• jjjj
• jjjjjj
• jjjjjj
• Bjjjj
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• Ajjjjj
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• PADPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDIN
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- vv[[[[[[[[[[[
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•
• PPPP
• ``YYYYYYYYYYY
• }YPPPPYYYYa
• YYYYYYYYYYYY
• YY`}
• YYYYY
• StrChrA
• StrPBrkA
• ntdll.dll
• MPR.dll
• SHLWAPI.dll
• StrCmpNW
• StrChrW
• StrCmpIW
• StrSpnA
• StrCpyNW
• StrCmpNIW
• StrStrIW
• CRYPT32.dll
• $@RSDS
• gz|z
• K_VI
• Hx]k
• syFbg
• .vbs
• {PC_ID}
• Sending stat %s, %s
• %[^/]%[/]%d
• %c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c
• "%s"
• encrypt
• /-Rb
• Mb=Lk
• NKagj(h
• !Q\t@
• t+@;
• UQPXY]Y[
• ;t$(v(
•]_^[
•]_^[
• t?xH
• QQQSVW
• QQVW
• YYPj@j
• QQVh
• hSv3_j
• hSv3_j
• hEX&vj;ht
• nj-h
• hSv3_j
• Vj\th
• hSv3_j
• QQQSVW
• QQQSVW
• QQQSVW
• QQQSVW

- QQQQSVW
- j Y+
- PhSv3_j
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
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- tSVW
- QZ^&
- Ph'c
- PhSv3_j
- $SVW
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- QQQQSVW
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- j\th@
- YYPh
- YYPj
- SVW
- ;HTw|j
- t?hT
- Phb$@
- crbr
- @tSj
- .rsrc

• Module 3 (probably unpacked / injected by the sample)

• Module 3 rich signatures

- 44616e530000000000000000000000000c30f0f0002000000c30f5f0001000000c30f5d00020000001b9dab000100000009788300030000000978930021000000000001007c0100001b9daf00510000001b9d9d00

• Module 3 yara matches

- GenerateTLSClientHelloPacket_Test ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- Str_Win32_Http_API (Match Windows Http API call)
- ASProtect_v132 ()
- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasDebugData (DebugData Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- Check_Dlls ()
- Check_Qemu_Description ()
- Check_Qemu_DeviceMap ()
- Check_VBox_Description ()

- Check_VBox_DeviceMap ()
- Check_VBox_Guest_Additions ()
- Check_VBox_VideoDrivers ()
- Check_VMWare_DeviceMap ()
- Check_VmTools ()
- Check_Wine ()
- vmdetect (Possibly employs anti-virtualization techniques)
- Check_OutputDebugStringA_iat ()
- anti_dbg (Checks if being debugged)
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_http (Communications over HTTP)
- escalate_priv (Escalade priviledges)
- win_mutex (Create or check mutex)
- win_registry (Affect system registries)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
- MD5_Constants (Look for MD5 constants)
- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

• Module 3 strings

• Module 3 most interesting strings

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache

- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer
- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI
- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA
- api | wininet: HttpSendRequestA
- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem
- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey
- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW

- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect
- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW
- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW
- api | advapi32: SetKernelObjectSecurity
- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree
- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId
- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA

- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW
- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW
- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW
- api | kernel32: CreateFileMappingW
- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary
- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW
- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW

- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW
- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode
- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

• Module 3 other strings

- Printers\\Defaults\\%s
- CERBER_KEY_PLACE
- NoDrives
- open
- EnableLUA
- /set {default} bootstatuspolicy ignoreallfailures
- /set {default} recoveryenabled no
- shadowcopy delete
- %s\\wbem\\wmic.exe
- delete shadows /all /quiet
- COMSPEC
- Installed
- PT0S
- PT1M
- 2000-01-01T00:00:00
- Component_02
- SeDebugPrivilege
- CERBER_BODY_PLACE
- CERBER_EVALUATED_CORE_PROTECTION_EVENT
- CERBER_CORE_PROTECTION_MUTEX
- \\x*x.exe
- update
- setup
- install
- sol.exe
- spider
- calc
- cmd.exe
- freecell
- mshearts

- ntbackup
- route
- task
- telnet
- winmine
- ping
- netsh
- notepad
- logoff
- reset
- msiexec
- winlogon
- disk
- lookup
- attrib
- find
- ftp.exe
- netstat
- write
- copy
- *.exe
- eshell32.dll
- "%s\\explorer.exe"
- %COMSPEC%
- runas
- Component_01
- Component_00
- files.txt
- Modified at %s %s
- ekernel32.dll
- vmmouse.sys
- SOFTWARE\\VMware, Inc.\\VMware Tools
- WMWARE
- VMWARE
- VBoxMouse.sys
- \\drivers\\
- SOFTWARE\\Oracle\\VirtualBox Guest Additions
- VIRTUALBOX
- VBOX
- QEMU
- HARDWARE\\DEVICEMAP\\Scsi\\Scsi Port 0\\Scsi Bus 0\\Target Id 0\\Logical Unit Id 0
- Identifier
- VideoBiosVersion
- PARALLELS
- HARDWARE\\Description\\System
- SystemBiosVersion
- multi_pot.exe
- hookexplorer.exe
- proc_analyzer.exe
- scktool.exe
- sniff_hit.exe
- sysanalyzer.exe
- idag.exe
- ollydbg.exe
- dumpcap.exe
- wireshark.exe
- \\\\.\\cv2k1

- \\sandbox\\
- \\sand-box\\
- Frz_State
- test_item.exe
- Ajjj
- Ajjj
- Ajjj
- U9f:f?6
- xo*2
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- A]YX
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- DyZE
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- PL4)n
- @S!E
- jF-i
- c9[$
- Ye5^
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- ZK21X
- wC4`
- Ehdi
- O9DV
- vz6y
- S2L!
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- AAG2
- uBxX
- q%y1n
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- FgE~
- c/5f5vc:
- 9a[%
- tZ0g
- fRXoD
- 6%&)
- D$ a
- D$ aX
- VerQueryValueW
- GetFileVersionInfoW
- StrToInt64ExA
- PathSkipRootW
- PathUnquoteSpacesW
- PathCombineW
- CoInitialize
- NETAPI32.dll
- OLEAUT32.dll
- SHELL32.dll
- ole32.dll
- USER32.dll

- ADVAPI32.dll
- KERNEL32.dll
- ShellExecuteW
- SHChangeNotify
- CoInitializeSecurity
- SHGetFolderPathW
- ShellExecuteA
- CoInitializeEx
- ShellExecuteExW
- CoUninitialize
- WS2_32.dll
- imagehlp.dll
- CheckSumMappedFile
- WNetEnumResourceW
- WNetOpenEnumW
- WNetCloseEnum
- VERSION.dll
- GetFileVersionInfoA
- GetFileVersionInfoSizeA
- VerQueryValueA
- GetFileVersionInfoSizeW
- PathMatchSpecW
- PathRemoveExtensionW
- StrCmpNIA
- PathFindFileNameW
- StrStrIA
- StrChrIA
- WININET.dll
- IA-*
- c"5o
- T1`m
- 4d(2d
- IDx3
- ?U Up
- Ydm3
- o38Fg$
- h6J-
- 4w%sd
- WV_i
- whitelist
- folders
- new_extension
- file_body
- file_extension
- files
- files_name
- help_files
- network
- debug
- activity
- language
- country
- vmware
- check
- countries
- property_name
- ip_geo

- languages
- blacklist
- kernel32.dll
- {COUNT_FILES}
- %02X%02X%02X%02X%02X%02X
- {MD5_KEY}
- {STOP_REASON}
- {OS}
- %05x
- {PARTNER_ID}
- {COUNTRY}
- {IS_X64}
- {IS_ADMIN}
- data_finish
- data_start
- knock
- port
- %02x
- timeout
- statistics
- servers
- %02X%02X%02X%02X%02X%02X%05X%03X
- %02x%02x%02x%02x%02x%02x%05xcerber
- open
- Stop reason: %s
- SHCreateItemFromParsingName
- requireAdministrator
- requestedExecutionLevel
- true
- autoElevate
- global_public_key
- autostart
- rsa_key_size
- min_file_size
- max_blocks
- max_block_size
- multithread
- close_process
- wine_get_unix_file_name
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- j8X+E
- j@X+E
- E8@P
- E8@P
- E(+E
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- u7j
- uYjY
- P8hT
- t6hH
- h<I@
- hjA@
- t5h`
- t0hT
- u:h|
- SVW2
- @.data
- `.rdata
- .text
- Rich
- !This program cannot be run in DOS mode.
- \\??\\
- *%S*
- %S%S
- /d /c taskkill /t /f /im "%s" > NUL & ping -n 1 127.0.0.1 > NUL & del "%s" > NUL
- "%s\\%s"
- shell.%s
- %s\\%S
- %s\\%s
- /d /c start "" "%s"
- Encrypting file %s...
- %s\\%s%S
- vmhgfs.sys
- VEN_%x
- \\\\.\\NPF_NdisWanIp
- \\cwsandbox\\
- dbghelp.dll
- dir_watch.dll
- sbiedll.dll
- jjjj
- jjjjjj
- jjjjjj
- Bjjjj
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- Ajjjjj
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- c@LSW
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- Fs&
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- pclq
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-
- PPPP
- ``YYYYYYYYYYY
- }YPPPPYYYYa
- YYYYYYYYYYYY
- YY`}
- YYYYY
- StrChrA
- StrPBrkA
- ntdll.dll
- MPR.dll
- SHLWAPI.dll
- StrCmpNW
- StrChrW
- StrCmpIW
- StrSpnA
- StrCpyNW

- StrCmpNIW
- StrStrIW
- CRYPT32.dll
- $@RSDS
- gz|z
- K_Vl
- Hx]k
- syFbg
- .vbs
- {PC_ID}
- Sending stat %s, %s
- %[^/]%[/]%d
- %c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c
- "%s"
- encrypt
- /-Rb
- Mb=Lk
- NKagj(h
- !Q\t@
- t+@;
- UQPXY]Y[
- ;t$(v(
-]_^[
-]_^[
- t?xH
- QQQSVW
- QQVW
- YYPj@j
- QQVh
- hSv3_j
- hSv3_j
- hEX&vj;ht
- nj-h
- hSv3_j
- Vj\th
- hSv3_j
- QQQSVW
- QQQSVW
- QQQSVW
- QQQSVW
- QQQQSVW
- j Y+
- PhSv3_j
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- @t&j;
- t&j;
- tSVW
- QZ^&
- Ph'c
- PhSv3_j
- $SVW
- j:Xf

- QQQQSVW
- YYh(
- j\th@
- YYPh
- YYPj
- SVW
- ;HTw|j
- t?hT
- Phb$@
- crbr
- @tSj
- .rsrc

• Module 4 (probably unpacked / injected by the sample)

• Module 4 rich signatures

- 44616e530000000000000000000000000c30f0f0002000000c30f5f0001000000c30f5d00020000001b9dab000100000000978830003000000009789300210000
00000001007c0100001b9daf00510000001b9d9d00

• Module 4 yara matches

- GenerateTLSClientHelloPacket_Test ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- Str_Win32_Http_API (Match Windows Http API call)
- ASProtect_v132 ()
- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasDebugData (DebugData Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- Check_Dlls ()
- Check_Qemu_Description ()
- Check_Qemu_DeviceMap ()
- Check_VBox_Description ()
- Check_VBox_DeviceMap ()
- Check_VBox_Guest_Additions ()
- Check_VBox_VideoDrivers ()
- Check_VMWare_DeviceMap ()
- Check_VmTools ()
- Check_Wine ()
- vmdetect (Possibly employs anti-virtualization techniques)
- Check_OutputDebugStringA_iat ()
- anti_dbg (Checks if being debugged)
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_http (Communications over HTTP)
- escalate_priv (Escalade priviledges)
- win_mutex (Create or check mutex)
- win_registry (Affect system registries)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)

- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
- MD5_Constants (Look for MD5 constants)
- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

• Module 4 strings

• Module 4 most interesting strings

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer

- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI
- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA
- api | wininet: HttpSendRequestA
- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem
- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey
- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW
- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect

- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW
- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW
- api | advapi32: SetKernelObjectSecurity
- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree
- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId
- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA
- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW

- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW
- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW
- api | kernel32: CreateFileMappingW
- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary
- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW
- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW
- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW

- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode
- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

• Module 4 other strings

- Printers\\Defaults\\%s
- CERBER_KEY_PLACE
- NoDrives
- open
- EnableLUA
- /set {default} bootstatuspolicy ignoreallfailures
- /set {default} recoveryenabled no
- shadowcopy delete
- %s\\wbem\\wmic.exe
- delete shadows /all /quiet
- COMSPEC
- Installed
- PT0S
- PT1M
- 2000-01-01T00:00:00
- Component_02
- SeDebugPrivilege
- CERBER_BODY_PLACE
- CERBER_EVALUATED_CORE_PROTECTION_EVENT
- CERBER_CORE_PROTECTION_MUTEX
- \\x*x.exe
- update
- setup
- install
- sol.exe
- spider
- calc
- cmd.exe
- freecell
- mshearts
- ntbackup
- route
- task
- telnet
- winmine
- ping
- netsh
- notepad
- logoff
- reset
- msiexec
- winlogon
- disk
- lookup
- attrib
- find
- ftp.exe

- netstat
- write
- copy
- *.exe
- eshell32.dll
- "%s\\explorer.exe"
- %COMSPEC%
- runas
- Component_01
- Component_00
- files.txt
- Modified at %s %s
- ekernel32.dll
- vmmouse.sys
- SOFTWARE\\VMware, Inc.\\VMware Tools
- WMWARE
- VMWARE
- VBoxMouse.sys
- \\drivers\\
- SOFTWARE\\Oracle\\VirtualBox Guest Additions
- VIRTUALBOX
- VBOX
- QEMU
- HARDWARE\\DEVICEMAP\\Scsi\\Scsi Port 0\\Scsi Bus 0\\Target Id 0\\Logical Unit Id 0
- Identifier
- VideoBiosVersion
- PARALLELS
- HARDWARE\\Description\\System
- SystemBiosVersion
- multi_pot.exe
- hookexplorer.exe
- proc_analyzer.exe
- scktool.exe
- sniff_hit.exe
- sysanalyzer.exe
- idag.exe
- ollydbg.exe
- dumpcap.exe
- wireshark.exe
- \\\\.\\cv2k1
- \\sandbox\\
- \\sand-box\\
- Frz_State
- test_item.exe
- Ajjj
- Ajjj
- Ajjj
- U9f:f?6
- xo*2
- _6}\\
- 6lpFn
- zj-o
- }I)E
- 8uI
- $@1'!
- ;
- AJ2u

- l9}i^
- .n=4H6
- oqJ=
- U*+3
- !O=B-om
- r|5U
- \\3x_
- Ysxa
- #b$52S
- #+3`
- /*2u
- kOeW:Q
- LHYId
- 09sr
- IM@;
- i&Sr;
- 'f1q
- ~1dqf
- *2,+
- lYuh
- uQkY
- -&PA;
- ^PAI
-]LHo
- iPGM
- fAjN#
- ed6T?H`:
- \\3AA3$
- H7OXJgM0
- Y9qaA
- z3yV
- }aRY
- [$~4
- g"2B
- #TeoD
- ibgUcS
- Uf[{
- TzU|
- M2)R
- 4yWwL
- aw-v
- AL`M
- jUdr
- g\\!go
- T{oeF{
- .D5
- DR-)I
- nUrxn
- H4%<
- ;93u
- _s!0P
- \\Re],
- +|1t
- :U∈
- %uB_
- a%qw
- \\1vk

- M!E)
- '^>ql
- safR
- b]^1
- P'!U
- G8Gf
- En5g>
- 1T">
- k~aC
- a2&)
- E+j
- M&1\\
- %gY;A-E
- -S6*m
- u;[c
- iDl8
- Zu8p
- (eoa
- O^WY
- >ZA@*
- V/nu
- [*1[
- KBAj
- s<{%3
- 8W!p
- 2~U`
- +"2t
- $2&,x
- oB1HG
- g2OZ#
- DD8V
- yi&py;
- ad&2
- }xozAf
- -% Qo
- b@t0
- `DII
- `EJ43
- \\ILD$
- N^o]
- %P3iP
- QN%nU
- d=*8
- l@?3
- -TQ5
- _Ml6Ze!
- K8zL~
- L=Uo
- Av0W
- #Q^89'
- 1qK#
- 6mV
- <M7o
- |efXa
- i*'U
- FE9C
- 4w&s;

- %1"Ip
- i9d{
- Z7]u
- AJZI#
- 7_%3
- *803E|RO
- e}b^&
-)OSH
- akw,
- ;w%5]
- n]8>
- *>PO
- 3}vH
- '5?f
- ``wOq7
- 5xXi
- %UaTv
- ,OvAWJ
- lCkO
-]KpB8
- 6icT
- Z3d.U
- ,xF9
- ik8I
-]V9N
- 2G;*.E
- eU;{FP3
- Ot\tf
- +cu5
- Om}
- 62Rg
- `25+'
- 5I-Ru
- O RoR
- Z14@
- 9u-T
- u#2
- //5@
- wQ5\\
- @i\\=
- Ux#1
- oIOn
- Z<in
- /&Iq;
- S-3[
- <$|Ki
- L5m=
- 0hj=
- A]YX
- 2z0`
- DyZE
- 8H="
- 5:[:
- PL4)n
- @S!E
- jF-i
- c9[$

- Ye5^
- 5#T\\
- 2tl\\
- `\\u:[
- ?oAS`
- [ce6
- &Sa4;
-]RXi
- ZK21X
- wC4`
- Ehdi
- O9DV
- vz6y
- S2L!
- 2,Uun~
- AAG2
- uBxX
- q%y1n
- WE&[ohT
- U`rik#
- FgE~
- c/5f5vc:
- 9a[%
- tZ0g
- fRXoD
- 6%&)
- D$ a
- D$ aX
- VerQueryValueW
- GetFileVersionInfoW
- StrToInt64ExA
- PathSkipRootW
- PathUnquoteSpacesW
- PathCombineW
- CoInitialize
- NETAPI32.dll
- OLEAUT32.dll
- SHELL32.dll
- ole32.dll
- USER32.dll
- ADVAPI32.dll
- KERNEL32.dll
- ShellExecuteW
- SHChangeNotify
- CoInitializeSecurity
- SHGetFolderPathW
- ShellExecuteA
- CoInitializeEx
- ShellExecuteExW
- CoUninitialize
- WS2_32.dll
- imagehlp.dll
- CheckSumMappedFile
- WNetEnumResourceW
- WNetOpenEnumW
- WNetCloseEnum
- VERSION.dll

- GetFileVersionInfoA
- GetFileVersionInfoSizeA
- VerQueryValueA
- GetFileVersionInfoSizeW
- PathMatchSpecW
- PathRemoveExtensionW
- StrCmpNIA
- PathFindFileNameW
- StrStrIA
- StrChrIA
- WININET.dll
- IA-*
- c"5o
- T1`m
- 4d(2d
- IDx3
- ?U Up
- Ydm3
- o38Fg$
- h6J-
- 4w%sd
- WV_i
- whitelist
- folders
- new_extension
- file_body
- file_extension
- files
- files_name
- help_files
- network
- debug
- activity
- language
- country
- vmware
- check
- countries
- property_name
- ip_geo
- languages
- blacklist
- kernel32.dll
- {COUNT_FILES}
- %02X%02X%02X%02X%02X%02X
- {MD5_KEY}
- {STOP_REASON}
- {OS}
- %05x
- {PARTNER_ID}
- {COUNTRY}
- {IS_X64}
- {IS_ADMIN}
- data_finish
- data_start
- knock
- port

- %02x
- timeout
- statistics
- servers
- %02X%02X%02X%02X%02X%02X%05X%03X
- %02x%02x%02x%02x%02x%02x%05xcerber
- open
- Stop reason: %s
- SHCreateItemFromParsingName
- requireAdministrator
- requestedExecutionLevel
- true
- autoElevate
- global_public_key
- autostart
- rsa_key_size
- min_file_size
- max_blocks
- max_block_size
- multithread
- close_process
- wine_get_unix_file_name
- Fcerber
- mSwO
- 6:fJ
- SVWUj
- tYVU
- SVWU
- VC20XC00U
- ,uej
- PhO?
- p hO?
- nosj
- 8nosjt
- j X+E
- j X+E
- rMaj
- ho>5aj
- j1h@
- Pha-A
- -uYh
- jxX+E
- j8X+E
- j@X+E
- E8@P
- E8@P
- E(+E
- E8@P
- hDo)
- h`5y
- u7j
- uYjY
- P8hT
- t6hH
- h<I@
- hjA@
- t5h`

- t0hT
- u:h|
- SVW2
- @.data
- `.rdata
- .text
- Rich
- !This program cannot be run in DOS mode.
- \\??\\
- *%S*
- %S%S
- /d /c taskkill /t /f /im "%s" > NUL & ping -n 1 127.0.0.1 > NUL & del "%s" > NUL
- "%s\\%s"
- shell.%s
- %s\\%S
- %s\\%s
- /d /c start "" "%s"
- Encrypting file %s...
- %s\\%s%S
- vmhgfs.sys
- VEN_%x
- \\\\.\\NPF_NdisWanIp
- \\cwsandbox\\
- dbghelp.dll
- dir_watch.dll
- sbiedll.dll
- jjjj
- jjjjjj
- jjjjjj
- Bjjjj
- jjjj
- jjjjj
- Ajjjjj
- jjjj
- jjjj
- jjjj
- PADPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADD
- w@!y*
- @+s@
- sxBG
- h>,@8},
- xzC%M
- \\-l`
- @GQS
- Lwz
- $!k=z
- hPt;\\=
- #&N;|c?Wl
- Kv/<
- [,D/'
- /S$M
- l$YS
- (WlxM
- Y.F(
- :~$`

- "-dV-
- :HgBx
- .YXn
- (>yl
- fQ(h
- -f\\dR
- S@Bz
- SxM)
- #//g
- @ #c
- r!y
- K/fcM)
- @z,H%
- h*(r
- (FrV
- =>Tv
- zsVbH&
- N+c
- xQRR
- gw\\K}i
- R(^F]
- SRi(}|
- F{t}$
- S{dg
- \\L/"
- ~% z
- /dn/
- Jy=Q
- /t>
- x*N>){
- kjxQ
- q<&n;<
- *\tm*
- gscd#
- JjcK
- y@'|
- d@y~f
- N;'F2%
- []^h
- qYS:
- PVkt
- -Qj/
- xTG(vb
- "J~/g
- dTYy
- \\X|gTb
- \\n>\\
- L?t-S:
- =Y}M
- <x"
- b\t-f
- t~WXs6
- @(t}
- }B#@=
- Vxg@
- ZXTq
- F,NR

- f[P&
- rH,G
- L+cX
- fqh!
- b>|l
- nrS8w\\
- c@LSW
- <W?prN
- }K-"
- MlD^
- khsT
- ^v*l
- /%dB
- P[G-
- c~'w
- \\xN/
- Of\\C*N}GS_a
- H*ZB
- 1|nd}td.
-]+lcam
- V~|g
- =HR`
- tcq@
- \\<W\tN
-]}JaQY
- *`\t~
- y)y[
- /vzYTWd
- ~Z@f
- t}+Sf
- %~$jG'
- $f}D
- ^{[B
- vV]x
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- P#{x
- vb*d
- $/mn
- bMZ\tW
- .{W[
- <j} l,
- mq+_p4"
- GYxc
- G%M@&
- /Rcy
- tDzd
- vdZq
- S_^W
- (Q,wDO
- mCSz
- R/wQF
- TCHyx
- q{==
- qPfj
- '!hMsa
- HjV
- * ^U{^

- jPZ.
- z"=q+++
- WgkQ
- Qp<p,
- Fs&
-]hSC
- BM"Yn
- %&"W
- pclq
- F-b[p1
- f?Mk
- /\t|y
- R:A@(v:K/
- DQJ:
- @@@@AI@@@@LB@@@@@@@@ODS@@@DWC\\@`@@@@@@@@@@@@@@dfnk@@jF@@DF@@[D@@
-
- PPPP
- vv[[[[[[[[[[[
- [[p[z
- [RRRR[[[[w|w
- [[[[[[[[[[[[]s
- d[[[[[
- JJJJKRJJJJOLJJJJJJJJUE@JJJEYMFJ]JJJJJJJJJJJJJacgNJJkmJJEmJJDEJJ
-
- PPPP
- ``YYYYYYYYYYY
- }YPPPPYYYYa
- YYYYYYYYYYYY
- YY`}
- YYYYY
- StrChrA
- StrPBrkA
- ntdll.dll
- MPR.dll
- SHLWAPI.dll
- StrCmpNW
- StrChrW
- StrCmpIW
- StrSpnA
- StrCpyNW
- StrCmpNIW
- StrStrIW
- CRYPT32.dll
- $@RSDS
- gz|z
- K_Vl
- Hx]k
- syFbg
- .vbs
- {PC_ID}
- Sending stat %s, %s
- %[^/]%[/]%d
- %c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c
- "%s"
- encrypt
- /-Rb
- Mb=Lk

- NKagj(h
- !Q\t@
- t+@;
- UQPXY]Y[
- ;t$(v(
-]_^[
-]_^[
- t?xH
- QQQSVW
- QQVW
- YYPj@j
- QQVh
- hSv3_j
- hSv3_j
- hEX&vj;ht
- nj-h
- hSv3_j
- Vj\th
- hSv3_j
- QQQSVW
- QQQSVW
- QQQSVW
- QQQSVW
- QQQQSVW
- j Y+
- PhSv3_j
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- QQQQSVW
- @t&j;
- t&j;
- tSVW
- QZ^&
- Ph'c
- PhSv3_j
- $SVW
- j:Xf
- QQQQSVW
- YYh(
- j\th@
- YYPh
- YYPj
- SVW
- ;HTw|j
- t?hT
- Phb$@
- crbr
- @tSj
- .rsrc

• Module 5 (probably unpacked / injected by the sample)

• Module 5 rich signatures

- 44616e530000000000000000000000000c30f0f0002000000c30f5f0001000000c30f5d00020000001b9dab00010000000097883000300000009789300210000
00000001007c0100001b9daf00510000001b9d9d00

• Module 5 yara matches

- GenerateTLSClientHelloPacket_Test ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- Str_Win32_Http_API (Match Windows Http API call)
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasDebugData (DebugData Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- Check_Dlls ()
- Check_Qemu_Description ()
- Check_Qemu_DeviceMap ()
- Check_VBox_Description ()
- Check_VBox_DeviceMap ()
- Check_VBox_Guest_Additions ()
- Check_VBox_VideoDrivers ()
- Check_VMWare_DeviceMap ()
- Check_VmTools ()
- Check_Wine ()
- vmdetect (Possibly employs anti-virtualization techniques)
- anti_dbg (Checks if being debugged)
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_http (Communications over HTTP)
- escalate_priv (Escalade priviledges)
- win_mutex (Create or check mutex)
- win_registry (Affect system registries)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)
- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
- MD5_Constants (Look for MD5 constants)
- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

• Module 5 strings

• Module 5 most interesting strings

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve

- path | path unit | wellknown extension: C:\\Users\\javi\\Downloads\\cebo.exe
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- path | path unit: C:\\Users\\javi\\Downloads\\
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer
- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI
- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA
- api | wininet: HttpSendRequestA
- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem

- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey
- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW
- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect
- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW
- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW
- api | advapi32: SetKernelObjectSecurity
- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree

- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId
- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA
- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW
- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW
- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW
- api | kernel32: CreateFileMappingW
- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary

- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW
- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW
- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW
- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode
- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

• Module 5 other strings

- shell.{76BF4271-6363-0293-2F1D-54E34C016282}
- Printers\\Defaults\\%s
- CERBER_KEY_PLACE
- NoDrives
- open
- EnableLUA
- /set {default} bootstatuspolicy ignoreallfailures

- /set {default} recoveryenabled no
- shadowcopy delete
- %s\\wbem\\wmic.exe
- delete shadows /all /quiet
- COMSPEC
- Installed
- PT0S
- PT1M
- 2000-01-01T00:00:00
- Component_02
- SeDebugPrivilege
- CERBER_BODY_PLACE
- CERBER_EVALUATED_CORE_PROTECTION_EVENT
- CERBER_CORE_PROTECTION_MUTEX
- \\x*x.exe
- update
- setup
- install
- sol.exe
- spider
- calc
- cmd.exe
- freecell
- mshearts
- ntbackup
- route
- task
- telnet
- winmine
- ping
- netsh
- notepad
- logoff
- reset
- msiexec
- winlogon
- disk
- lookup
- attrib
- find
- ftp.exe
- netstat
- write
- copy
- *.exe
- eshell32.dll
- "%s\\explorer.exe"
- %COMSPEC%
- runas
- Component_01
- Component_00
- files.txt
- Modified at %s %s
- ekernel32.dll
- vmmouse.sys
- SOFTWARE\\VMware, Inc.\\VMware Tools
- WMWARE

- VMWARE
- VBoxMouse.sys
- \\drivers\\
- SOFTWARE\\Oracle\\VirtualBox Guest Additions
- VIRTUALBOX
- VBOX
- QEMU
- HARDWARE\\DEVICEMAP\\Scsi\\Scsi Port 0\\Scsi Bus 0\\Target Id 0\\Logical Unit Id 0
- Identifier
- VideoBiosVersion
- PARALLELS
- HARDWARE\\Description\\System
- SystemBiosVersion
- multi_pot.exe
- hookexplorer.exe
- proc_analyzer.exe
- scktool.exe
- sniff_hit.exe
- sysanalyzer.exe
- idag.exe
- ollydbg.exe
- dumpcap.exe
- wireshark.exe
- \\\\.\\cv2k1
- \\sandbox\\
- \\sand-box\\
- Frz_State
- test_item.exe
- Ajjj
- Ajjj
- Ajjj
- U9f:f?6
- xo*2
- _6}\\
- 6lpFn
- zj-o
- }!)E
- 8uI
- $@1'!
- ;
- AJ2u
- l9}j^
- .n=4H6
- oqJ=
- U*+3
- !O=B-om
- r|5U
- \\3x_
- Ysxa
- #b$52S
- #+3`
- /*2u
- kOeW:Q
- LHYId
- 09sr
- IM@;
- i&Sr;

- 'f1q
- ~1dqf
- *2,+
- lYuh
- uQkY
- -&PA;
- ^PAI
-]LHo
- iPGM
- fAjN#
- ed6T?H`:
- \\3AA3$
- H7OXJgM0
- Y9qaA
- z3yV
- }aRY
- [$~4
- g"2B
- #TeoD
- ibgUcS
- Uf[{
- TzU|
- M2)R
- 4yWwL
- aw-v
- AL`M
- jUdr
- g\\!go
- T{oeF{
- .D5
- DR-)I
- nUrxn
- H4%<
- ;93u
- _s!0P
- \\Re],
- +|1t
- :U∈
- %uB_
- a%qw
- \\1vk
- M!E)
- '^>ql
- safR
- b]^1
- P'!U
- G8Gf
- En5g>
- 1T">
- k~aC
- a2&)
- E+j
- M&1\\
- %gY;A-E
- -S6*m
- u;[c
- iDl8

- Zu8p
- (eoa
- O^WY
- >ZA@*
- V/nu
- [*1[
- KBAj
- s<{%3
- 8W!p
- 2~U`
- +"2t
- $2&,x
- oB1HG
- g2OZ#
- DD8V
- yi&py;
- ad&2
- }xozAf
- -% Qo
- b@t0
- `DII
- `EJ43
- \\ILD$
- N^o]
- %P3iP
- QN%nU
- d=*8
- l@?3
- -TQ5
- _Ml6Ze!
- K8zL~
- L=Uo
- Av0W
- #Q^89'
- 1qK#
- 6mV
- <M7o
- |efXa
- i*'U
- FE9C
- 4w&s;
- %1"Ip
- i9d{
- Z7]u
- AJZI#
- 7_%3
- *803E|RO
- e}b^&
-)OSH
- akw,
- ;w%5]
- n]8>
- *>PO
- 3}vH
- '5?f
- ``wOq7
- 5xXi

- %UaTv
- ,OvAWJ
- lCkO
-]KpB8
- 6icT
- Z3d.U
- ,xF9
- ik8I
-]V9N
- 2G;*.E
- eU;{FP3
- Ot\tf
- +cu5
- Om}
- 62Rg
- `25+'
- 5I-Ru
- O RoR
- Z14@
- 9u-T
- u#2
- //5@
- wQ5\\
- @i\\=
- Ux#1
- oIOn
- Z<in
- /&Iq;
- S-3[
- <$|Ki
- L5m=
- 0hj=
- A]YX
- 2z0`
- DyZE
- 8H="
- 5:[:
- PL4)n
- @S!E
- jF-i
- c9[$
- Ye5^
- 5#T\\
- 2tl\\
- `\\u:[
- ?oAS`
- [ce6
- &Sa4;
-]RXi
- ZK21X
- wC4`
- Ehdi
- O9DV
- vz6y
- S2L!
- 2,Uun~
- AAG2

- uBxX
- q%y1n
- WE&[ohT
- U`rik#
- FgE~
- c/5f5vc:
- 9a[%
- tZ0g
- fRXoD
- 6%&)
- vL8l
- >7O-
- D$ a
- D$ aX
- VerQueryValueW
- GetFileVersionInfoW
- StrToInt64ExA
- PathSkipRootW
- PathUnquoteSpacesW
- PathCombineW
- CoInitialize
- NETAPI32.dll
- OLEAUT32.dll
- SHELL32.dll
- ole32.dll
- USER32.dll
- ADVAPI32.dll
- KERNEL32.dll
- ShellExecuteW
- SHChangeNotify
- CoInitializeSecurity
- SHGetFolderPathW
- ShellExecuteA
- CoInitializeEx
- ShellExecuteExW
- CoUninitialize
- WS2_32.dll
- imagehlp.dll
- CheckSumMappedFile
- WNetEnumResourceW
- WNetOpenEnumW
- WNetCloseEnum
- VERSION.dll
- GetFileVersionInfoA
- GetFileVersionInfoSizeA
- VerQueryValueA
- GetFileVersionInfoSizeW
- PathMatchSpecW
- PathRemoveExtensionW
- StrCmpNIA
- PathFindFileNameW
- StrStrIA
- StrChrIA
- WININET.dll
- lA-*
- c"5o
- T1`m

- 4d(2d
- IDx3
- ?U Up
- Ydm3
- o38Fg$
- h6J-
- 4w%sd
- WV_i
- whitelist
- folders
- new_extension
- file_body
- file_extension
- files
- files_name
- help_files
- network
- debug
- activity
- language
- country
- vmware
- check
- countries
- property_name
- ip_geo
- languages
- blacklist
- kernel32.dll
- {COUNT_FILES}
- %02X%02X%02X%02X%02X%02X
- {MD5_KEY}
- {STOP_REASON}
- {OS}
- %05x
- {PARTNER_ID}
- {COUNTRY}
- {IS_X64}
- {IS_ADMIN}
- data_finish
- data_start
- knock
- port
- %02x
- timeout
- statistics
- servers
- %02X%02X%02X%02X%02X%02X%05X%03X
- %02x%02x%02x%02x%02x%02x%05xcerber
- open
- Stop reason: %s
- SHCreateItemFromParsingName
- requireAdministrator
- requestedExecutionLevel
- true
- autoElevate
- global_public_key

- autostart
- rsa_key_size
- min_file_size
- max_blocks
- max_block_size
- multithread
- close_process
- wine_get_unix_file_name
- Fcerber
- mSwO
- 6:fJ
- EG>v
- Y-vA
- 3.vw
- 5.vcv.v6
- ,va>-v1#-v
- u.vO!.v
- 3.vQJ.vo
- SVWUj
- tYVU
- SVWU
- VC20XC00U
- ,uej
- PhO?
- p hO?
- nosj
- 8nosjt
- j X+E
- j X+E
- rMaj
- ho>5aj
- j1h@
- Pha-A
- -uYh
- jxX+E
- j8X+E
- j@X+E
- E8@P
- E8@P
- E(+E
- E8@P
- hDo)
- h`5y
- u7j
- uYjY
- P8hT
- t6hH
- h<I@
- hjA@
- t5h`
- t0hT
- u:h|
- SVW2
- @.data
- `.rdata
- .text
- Rich

- !This program cannot be run in DOS mode.
- \\??\\
- *%S*
- %S%S
- /d /c taskkill /t /f /im "%s" > NUL & ping -n 1 127.0.0.1 > NUL & del "%s" > NUL
- "%s\\%s"
- shell.%s
- %s\\%S
- %s\\%s
- /d /c start "" "%s"
- Encrypting file %s...
- %s\\%s%S
- vmhgfs.sys
- VEN_%x
- \\\\.\\NPF_NdisWanIp
- \\cwsandbox\\
- dbghelp.dll
- dir_watch.dll
- sbiedll.dll
- jjjj
- jjjjjj
- jjjjjj
- Bjjjj
- jjjj
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- Ajjjjj
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- PADPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADDINGXXPADDINGPADD
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- YY`}
- YYYYY
- StrChrA
- StrPBrkA
- ntdll.dll
- MPR.dll
- SHLWAPI.dll
- StrCmpNW
- StrChrW
- StrCmpIW
- StrSpnA
- StrCpyNW
- StrCmpNIW
- StrStrIW
- CRYPT32.dll
- $@RSDS
- gz|z
- K_Vl
- Hx]k
- syFbg
- .vbs
- {PC_ID}
- Sending stat %s, %s
- %[^/]%[/]%d
- %c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c
- "%s"
- encrypt
- /-Rb
- Mb=Lk
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- YYPh
- YYPj
- SVW
- ;HTw|j
- t?hT
- Phb$@
- crbr
- @tSj
- .rsrc

• Module 6 (probably unpacked / injected by the sample)

• Module 6 rich signatures

- 44616e530000000000000000000000000c30f0f0002000000c30f5f0001000000c30f5d00020000001b9dab000100000000978830003000000009789300210000
00000001007c0100001b9daf00510000001b9d9d00

• Module 6 yara matches

- GenerateTLSClientHelloPacket_Test ()
- Str_Win32_Winsock2_Library (Match Winsock 2 API library declaration)
- Str_Win32_Wininet_Library (Match Windows Inet API library declaration)
- Str_Win32_Internet_API (Match Windows Inet API call)
- Str_Win32_Http_API (Match Windows Http API call)
- ASProtect_v132 ()
- Microsoft_Visual_Cpp_v50v60_MFC ()
- IsPE32 ()
- IsWindowsGUI ()
- HasOverlay (Overlay Check)
- HasDebugData (DebugData Check)
- HasRichSignature (Rich Signature Check)
- DebuggerCheck__QueryInfo ()
- Check_Dlls ()
- Check_Qemu_Description ()
- Check_Qemu_DeviceMap ()
- Check_VBox_Description ()
- Check_VBox_DeviceMap ()
- Check_VBox_Guest_Additions ()
- Check_VBox_VideoDrivers ()
- Check_VMWare_DeviceMap ()
- Check_VmTools ()
- Check_Wine ()
- vmdetect (Possibly employs anti-virtualization techniques)
- Check_OutputDebugStringA_iat ()
- anti_dbg (Checks if being debugged)
- inject_thread (Code injection with CreateRemoteThread in a remote process)
- network_http (Communications over HTTP)
- escalate_priv (Escalade priviledges)
- win_mutex (Create or check mutex)
- win_registry (Affect system registries)
- win_token (Affect system token)
- win_files_operation (Affect private profile)
- vmdetect_misc (Following Rule is referenced from AlienVault\\'s Yara rule repository.This rule contains additional processes and driver names.)
- without_attachments (Rule to detect the no presence of any attachment)
- without_images (Rule to detect the no presence of any image)

- without_urls (Rule to detect the no presence of any url)
- contentis_base64 (This rule finds for base64 strings)
- Crypt32_CryptBinaryToString_API (Looks for crypt32 CryptBinaryToStringA function)
- MD5_Constants (Look for MD5 constants)
- RIPEMD160_Constants (Look for RIPEMD-160 constants)
- SHA1_Constants (Look for SHA1 constants)
- Prime_Constants_long (List of primes [long])

• Module 6 strings

• Module 6 most interesting strings

- suspicious event or string: bcdedit.exe
- suspicious event or string: %s\\vssadmin.exe
- url | valid domains | tlds: T.lU
- url | valid domains | tlds: r.Ve
- path | path unit | wellknown extension: C:\\test\\cerber_debug.txt
- path | path unit | wellknown extension: C:\\TOOLS\\execute.exe
- path | path unit | wellknown extension: C:\\stimulator.exe
- path | path unit | wellknown extension: C:\\popupkiller.exe
- api | frequently used by malware: WideCharToMultiByte
- api | frequently used by malware: ZwOpenSection
- api | frequently used by malware: AdjustTokenPrivileges
- api | frequently used by malware: CreateToolhelp32Snapshot
- api | frequently used by malware: ZwOpenProcess
- api | frequently used by malware: OpenProcess
- api | frequently used by malware: ZwQuerySystemInformation
- api | frequently used by malware: OpenProcessToken
- api | frequently used by malware: GetNativeSystemInfo
- api | frequently used by malware: GetTickCount
- api | frequently used by malware: TranslateMessage
- api | frequently used by malware: ResumeThread
- api | frequently used by malware: WriteProcessMemory
- api | frequently used by malware: WinExec
- api | frequently used by malware: ZwQueryInformationProcess
- api | frequently used by malware: VirtualAllocEx
- api | frequently used by malware: FlushInstructionCache
- api | frequently used by malware: ReadProcessMemory
- api | frequently used by malware: MapViewOfFile
- api | frequently used by malware: CoCreateInstance
- api | frequently used by malware: GetForegroundWindow
- api | frequently used by malware: ExitProcess
- api | frequently used by malware: CryptEncrypt
- api | frequently used by malware: CreateThread
- api | frequently used by malware: Sleep
- api | frequently used by malware: WriteFile
- api | frequently used by malware: SetFileTime
- api | frequently used by malware: GetProcAddress
- api | frequently used by malware: RegCloseKey
- api | frequently used by malware: LoadResource
- api | frequently used by malware: InternetCloseHandle
- api | frequently used by malware: InternetReadFile
- api | frequently used by malware: send_stat
- registry: Software\\Microsoft\\Windows\\CurrentVersion\\Policies\\Explorer
- registry: SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Policies\\System
- registry: SYSTEM\\CurrentControlSet\\Enum\\PCI

- path | path unit: F:\\trash\\code\\work\\cerber\\bin\\Debug\\cerber_x86.pdb
- api | crypt32: CryptBinaryToStringA
- api | crypt32: CryptStringToBinaryA
- api | wininet: InternetOpenA
- api | wininet: HttpSendRequestA
- api | wininet: HttpOpenRequestA
- api | wininet: InternetCrackUrlA
- api | wininet: InternetConnectA
- api | crypt32: CryptImportPublicKeyInfo
- api | crypt32: CryptDecodeObjectEx
- api | ntdll: api_log.dll
- api | ntdll: _alldiv
- api | ntdll: _chkstk
- api | ntdll: _allmul
- api | ntdll: _aullrem
- api | ntdll: _aulldiv
- api | ntdll: _aullshr
- api | ntdll: _allshl
- api | ntdll: NtQueryVirtualMemory
- api | kernel32: RtlUnwind
- api | kernel32: RegFlushKey
- api | kernel32: RegEnumValueW
- api | kernel32: RegDeleteValueW
- api | ntdll: isspace
- api | kernel32: WaitForMultipleObjects
- api | kernel32: GetCurrentThreadId
- api | kernel32: IsBadStringPtrA
- api | kernel32: IsBadCodePtr
- api | kernel32: IsBadReadPtr
- api | kernel32: IsBadStringPtrW
- api | user32: UnregisterClassW
- api | user32: RegisterClassExW
- api | advapi32: CreateWellKnownSid
- api | advapi32: CheckTokenMembership
- api | advapi32: FreeSid
- api | advapi32: AllocateAndInitializeSid
- api | advapi32: DuplicateToken
- api | kernel32: CreateDirectoryW
- api | kernel32: ExpandEnvironmentStringsW
- api | advapi32: ConvertSidToStringSidW
- api | advapi32: GetLengthSid
- api | netapi: NetUserEnum
- api | netapi: NetApiBufferFree
- api | netapi: NetUserGetInfo
- api | kernel32: TlsAlloc
- api | kernel32: IsBadWritePtr
- api | kernel32: lstrcpynA
- api | kernel32: MultiByteToWideChar
- api | ntdll: memmove
- api | ntdll: RtlDosPathNameToNtPathName_U
- api | ntdll: NtDeleteFile
- api | ntdll: RtlFreeUnicodeString
- api | kernel32: SearchPathW
- api | kernel32: DuplicateHandle
- api | kernel32: VirtualProtect
- api | kernel32: GetModuleHandleA
- api | kernel32: lstrcmpiW

- api | kernel32: Process32NextW
- api | kernel32: GetSystemInfo
- api | advapi32: ConvertStringSecurityDescriptorToSecurityDescriptorW
- api | kernel32: Process32FirstW
- api | advapi32: SetKernelObjectSecurity
- api | kernel32: GetHandleInformation
- api | ntdll: ZwClose
- api | advapi32: LookupPrivilegeValueW
- api | ntdll: ZwOpenDirectoryObject
- api | user32: GetLastInputInfo
- api | advapi32: GetTokenInformation
- api | kernel32: GetComputerNameA
- api | ntdll: memcpy
- api | ntdll: memset
- api | kernel32: HeapFree
- api | kernel32: HeapValidate
- api | kernel32: HeapReAlloc
- api | kernel32: GetProcessHeaps
- api | kernel32: HeapCreate
- api | kernel32: HeapSetInformation
- api | kernel32: GetCurrentProcessId
- api | kernel32: HeapAlloc
- api | kernel32: SetLastError
- api | kernel32: RegSetValueExW
- api | kernel32: FindNextFileW
- api | kernel32: QueryDosDeviceW
- api | kernel32: VirtualAlloc
- api | kernel32: VirtualFree
- api | kernel32: GetLogicalDrives
- api | kernel32: GetDriveTypeW
- api | kernel32: GetModuleFileNameA
- api | advapi32: RegOpenKeyW
- api | kernel32: lstrcmpiA
- api | kernel32: GetVersionExW
- api | kernel32: RegCreateKeyExW
- api | user32: GetKeyboardLayoutList
- api | kernel32: GetEnvironmentVariableW
- api | kernel32: lstrcpyA
- api | user32: GetSystemMetrics
- api | kernel32: lstrlenA
- api | user32: CharLowerBuffA
- api | kernel32: SetProcessShutdownParameters
- api | kernel32: OutputDebugStringA
- api | kernel32: SetCurrentDirectoryW
- api | user32: wsprintfA
- api | kernel32: SetThreadPriority
- api | user32: PeekMessageW
- api | kernel32: GetCurrentThread
- api | user32: CreateWindowExW
- api | kernel32: GetCurrentProcess
- api | user32: RegisterClassW
- api | kernel32: CreateMutexW
- api | user32: DefWindowProcW
- api | user32: DispatchMessageW
- api | kernel32: SetFileAttributesW
- api | kernel32: FindClose
- api | kernel32: GetModuleFileNameW

- api | kernel32: FindFirstFileW
- api | kernel32: DeleteFileW
- api | kernel32: GetWindowsDirectoryW
- api | kernel32: OpenEventW
- api | kernel32: CreateFileMappingW
- api | kernel32: GetTempPathW
- api | kernel32: FlushFileBuffers
- api | kernel32: TerminateProcess
- api | kernel32: lstrcpynW
- api | kernel32: GetSystemWow64DirectoryW
- api | kernel32: CopyFileW
- api | kernel32: LoadLibraryW
- api | kernel32: LoadLibraryExW
- api | kernel32: CreateProcessW
- api | kernel32: FreeLibrary
- api | kernel32: UnmapViewOfFile
- api | kernel32: GetFileSize
- api | kernel32: GetTempFileNameW
- api | kernel32: LocalFree
- api | advapi32: CryptGetKeyParam
- api | advapi32: CryptAcquireContextW
- api | kernel32: lstrcpyW
- api | kernel32: FileTimeToLocalFileTime
- api | kernel32: CloseHandle
- api | kernel32: DeleteCriticalSection
- api | kernel32: GetFileTime
- api | advapi32: CryptDestroyKey
- api | kernel32: CreateEventW
- api | kernel32: EnterCriticalSection
- api | kernel32: MoveFileW
- api | kernel32: GetFileSizeEx
- api | kernel32: ReadFile
- api | kernel32: FileTimeToSystemTime
- api | kernel32: GetFileAttributesW
- api | kernel32: GetTimeFormatW
- api | kernel32: LeaveCriticalSection
- api | kernel32: InitializeCriticalSection
- api | kernel32: OutputDebugStringW
- api | kernel32: SetEvent
- api | kernel32: WaitForSingleObject
- api | kernel32: SetFilePointerEx
- api | kernel32: SetFilePointer
- api | kernel32: GetDateFormatW
- api | kernel32: lstrlenW
- api | kernel32: lstrcatW
- api | kernel32: GetSystemDirectoryW
- api | user32: wsprintfW
- api | advapi32: RegEnumKeyW
- api | kernel32: RegOpenKeyExW
- api | kernel32: RegQueryValueExW
- api | kernel32: GetVolumeInformationW
- api | kernel32: GetLastError
- api | kernel32: OpenMutexW
- api | kernel32: CreateFileW
- api | kernel32: GetModuleHandleW
- api | kernel32: GetSystemWindowsDirectoryW
- api | kernel32: SetErrorMode

- api | kernel32: LockResource
- api | kernel32: SizeofResource
- api | kernel32: FindResourceW
- api | kernel32: FreeResource

• Module 6 other strings

- Printers\\Defaults\\%s
- CERBER_KEY_PLACE
- NoDrives
- open
- EnableLUA
- /set {default} bootstatuspolicy ignoreallfailures
- /set {default} recoveryenabled no
- shadowcopy delete
- %s\\wbem\\wmic.exe
- delete shadows /all /quiet
- COMSPEC
- Installed
- PT0S
- PT1M
- 2000-01-01T00:00:00
- Component_02
- SeDebugPrivilege
- CERBER_BODY_PLACE
- CERBER_EVALUATED_CORE_PROTECTION_EVENT
- CERBER_CORE_PROTECTION_MUTEX
- \\x*x.exe
- update
- setup
- install
- sol.exe
- spider
- calc
- cmd.exe
- freecell
- mshearts
- ntbackup
- route
- task
- telnet
- winmine
- ping
- netsh
- notepad
- logoff
- reset
- msiexec
- winlogon
- disk
- lookup
- attrib
- find
- ftp.exe
- netstat
- write

- copy
- *.exe
- eshell32.dll
- "%s\\explorer.exe"
- %COMSPEC%
- runas
- Component_01
- Component_00
- files.txt
- Modified at %s %s
- ekernel32.dll
- vmmouse.sys
- SOFTWARE\\VMware, Inc.\\VMware Tools
- WMWARE
- VMWARE
- VBoxMouse.sys
- \\drivers\\
- SOFTWARE\\Oracle\\VirtualBox Guest Additions
- VIRTUALBOX
- VBOX
- QEMU
- HARDWARE\\DEVICEMAP\\Scsi\\Scsi Port 0\\Scsi Bus 0\\Target Id 0\\Logical Unit Id 0
- Identifier
- VideoBiosVersion
- PARALLELS
- HARDWARE\\Description\\System
- SystemBiosVersion
- multi_pot.exe
- hookexplorer.exe
- proc_analyzer.exe
- scktool.exe
- sniff_hit.exe
- sysanalyzer.exe
- idag.exe
- ollydbg.exe
- dumpcap.exe
- wireshark.exe
- \\\\.\\cv2k1
- \\sandbox\\
- \\sand-box\\
- Frz_State
- test_item.exe
- Ajjj
- Ajjj
- Ajjj
- U9f:f?6
- xo*2
- _6}\\
- 6lpFn
- zj-o
- }!)E
- 8uI
- $@1'!
- ;
- AJ2u
- l9}j^
- .n=4H6

- oqJ=
- U*+3
- !O=B-om
- r|5U
- \\3x_
- Ysxa
- #b$52S
- #+3`
- /*2u
- kOeW:Q
- LHYId
- 09sr
- IM@;
- i&Sr;
- 'f1q
- ~1dqf
- *2,+
- lYuh
- uQkY
- -&PA;
- ^PAI
-]LHo
- iPGM
- fAjN#
- ed6T?H`:
- \\3AA3$
- H7OXJgM0
- Y9qaA
- z3yV
- }aRY
- [$~4
- g"2B
- #TeoD
- ibgUcS
- Uf[{
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- `EJ43
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- QN%nU
- d=*8
- l@?3
- -TQ5
- _Ml6Ze!
- K8zL~
- L=Uo
- Av0W
- #Q^89'
- 1qK#
- 6mV
- <M7o
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- i*'U
- FE9C
- 4w&s;
- %1"Ip
- i9d{

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- ``wOq7
- 5xXi
- %UaTv
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- lCkO
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- ik8I
-]V9N
- 2G;*.E
- eU;{FP3
- Ot\tf
- +cu5
- Om}
- 62Rg
- `25+'
- 5I-Ru
- O RoR
- Z14@
- 9u-T
- u#2
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- 0hj=
- A]YX
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- Ye5^
- 5#T\\

- 2tl\\
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- &Sa4;
-]RXi
- ZK21X
- wC4`
- Ehdi
- O9DV
- vz6y
- S2L!
- 2,Uun~
- AAG2
- uBxX
- q%y1n
- WE&[ohT
- U`rik#
- FgE~
- c/5f5vc:
- 9a[%
- tZ0g
- fRXoD
- 6%&)
- D$ a
- D$ aX
- VerQueryValueW
- GetFileVersionInfoW
- StrToInt64ExA
- PathSkipRootW
- PathUnquoteSpacesW
- PathCombineW
- CoInitialize
- NETAPI32.dll
- OLEAUT32.dll
- SHELL32.dll
- ole32.dll
- USER32.dll
- ADVAPI32.dll
- KERNEL32.dll
- ShellExecuteW
- SHChangeNotify
- CoInitializeSecurity
- SHGetFolderPathW
- ShellExecuteA
- CoInitializeEx
- ShellExecuteExW
- CoUninitialize
- WS2_32.dll
- imagehlp.dll
- CheckSumMappedFile
- WNetEnumResourceW
- WNetOpenEnumW
- WNetCloseEnum
- VERSION.dll
- GetFileVersionInfoA
- GetFileVersionInfoSizeA

- VerQueryValueA
- GetFileVersionInfoSizeW
- PathMatchSpecW
- PathRemoveExtensionW
- StrCmpNIA
- PathFindFileNameW
- StrStrIA
- StrChrIA
- WININET.dll
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- T1`m
- 4d(2d
- lDx3
- ?U Up
- Ydm3
- o38Fg$
- h6J-
- 4w%sd
- WV_i
- whitelist
- folders
- new_extension
- file_body
- file_extension
- files
- files_name
- help_files
- network
- debug
- activity
- language
- country
- vmware
- check
- countries
- property_name
- ip_geo
- languages
- blacklist
- kernel32.dll
- {COUNT_FILES}
- %02X%02X%02X%02X%02X%02X
- {MD5_KEY}
- {STOP_REASON}
- {OS}
- %05x
- {PARTNER_ID}
- {COUNTRY}
- {IS_X64}
- {IS_ADMIN}
- data_finish
- data_start
- knock
- port
- %02x
- timeout

- statistics
- servers
- %02X%02X%02X%02X%02X%02X%05X%03X
- %02x%02x%02x%02x%02x%02x%05xcerber
- open
- Stop reason: %s
- SHCreateItemFromParsingName
- requireAdministrator
- requestedExecutionLevel
- true
- autoElevate
- global_public_key
- autostart
- rsa_key_size
- min_file_size
- max_blocks
- max_block_size
- multithread
- close_process
- wine_get_unix_file_name
- Fcerber
- mSwO
- 6:fJ
- SVWUj
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- SVWU
- VC20XC00U
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- P8hT
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- SVW2
- @.data
- `.rdata
- .text
- Rich
- !This program cannot be run in DOS mode.
- \\??\\
- *%S*
- %S%S
- /d /c taskkill /t /f /im "%s" > NUL & ping -n 1 127.0.0.1 > NUL & del "%s" > NUL
- "%s\\%s"
- shell.%s
- %s\\%S
- %s\\%s
- /d /c start "" "%s"
- Encrypting file %s...
- %s\\%s%S
- vmhgfs.sys
- VEN_%x
- \\\\.\\NPF_NdisWanIp
- \\cwsandbox\\
- dbghelp.dll
- dir_watch.dll
- sbiedll.dll
- jjjj
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- Bjjjj
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- YY`}
- YYYYY
- StrChrA
- StrPBrkA
- ntdll.dll
- MPR.dll
- SHLWAPI.dll
- StrCmpNW
- StrChrW
- StrCmpIW
- StrSpnA
- StrCpyNW
- StrCmpNIW
- StrStrIW
- CRYPT32.dll
- $@RSDS
- gz|z
- K_VI
- Hx]k
- syFbg
- .vbs
- {PC_ID}
- Sending stat %s, %s
- %[^/]%[/]%d
- %c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c-%c%c%c%c
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- encrypt
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